



NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008

Department	Computer Science & Engineering
Examination	Mid Semester
Program	B.Tech/PhD
Session	Spring
Course code & Course Name	CS6510 (Deep Learning)
Marks	30
Duration	2 hours

There are ... 5 ... questions printed on ...02... pages

Instructions: 1. In case of any doubt, write your assumptions, write it clearly and continue.
2. There are 05 questions printed on two pages of the paper. Attempt all the questions.

Q. No.	Questions	Marks	CO*	BL**
1. (A)	Explain the McCulloch-Pitts (MP) neuron along with its limitations. How the perceptron model overcomes the limitations of the MP neuron?	[3]	CO1	Remembering (L6) Understanding (L5)
(B)	If the sets P and N are finite and linearly separable, prove that the perceptron learning algorithm updates the weight vector w_t a finite number of times such that the two sets are separated.	[3]		
2. (A)	Determine whether it is possible to implement the XOR gate logic using an MP-neuron. What adjustments are necessary for the implementation in the above scenario?	[2]	CO1	Understanding (L5) Applying (L4)
(B)	Derive the mathematical formulation of gradient descent and explain its working with an example. Show that the Gradient Descent will converge in finite steps.	[4]		
3.	Explain the phenomenon of backpropagation for the thin ANN as described in the below figure. State the equations in forward pass and equation for weight updation for all the layers. The loss function in the hidden layer and output layer is sigmoid. Assume the loss function as binary cross entropy and learning rate as η.	[6]	CO1	Understanding (L5) Applying (L4)

4.	Derive the updated rules for the following variants of Gradient Descent A. Nesterov accelerated GD. B. RMSProp C. Adam	[2] [2] [2]	CO2	Understanding (L5) Applying (L4) Analyzing (L3)
5. (A)	Derive the numerical expression of the update rule adding L2 regularization and explain how it affects model weights update over multiple training epochs.	[3]	CO2	Understanding (L5) Applying (L4) Creativity (L1)
(B)	Explain three regularization methods to overcome the issue of overfitting of a complex model.	[3]		

CO**: Course Outcome; *BL**: Bloom's Taxonomy Level.